



Perspective

An Atlas of optical skyrmions: charting topologically equivalent field configurations

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Abstract

Optical skyrmion fields inherit the exceptional dynamic shaping capabilities of structured light. This distinguishes them from their counterparts in other physical systems, yet ongoing research remains largely centred on well-known textures such as Néel-type skyrmions, Bloch-type skyrmions and bimerons. We chart topologically equivalent optical skyrmion fields and discuss how this property could potentially be used for communication applications. We previously showed that Stokes phase plots are effective tools to characterize skyrmion fields experimentally, we develop this theory and examine skyrmion beams through the lens of constellations of polarization singularities.

Keywords: skyrmions, Poincaré beams, structured light, topological optics

1. Introduction

Topological invariants have become a focal point in photonics research as they offer robust pathways for communication and sensing. Robust against continuous space deformations, these integer-valued quantities have been studied extensively in the context of singular optics where they quantify the phase variation around an optical vortex or the variation in polarization orientation around a polarization singularity [1].

The emergence of paraxial skyrmion beams as novel polarization textures of light [2–5] introduced a novel topological invariant into singular optics, namely, the skyrmion number quantifying the coverage of the Poincaré sphere, with exciting prospects for information processing and storage [6, 7]. Skyrmions as topological objects can be found across a wide range of disciplines, including nuclear and atomic physics [8, 9], magnetism [10], acoustics [11] and more recently in water-wave systems [12]. They can also take on many forms, and have already been observed in optics such as Néel-type and Bloch-type skyrmions [13], bimerons [14] and multi-skyrmions [15].

While considerable emphasis is currently being placed on these particular field configurations, state-of-the-art light structuring techniques grant access to a much larger library of skyrmion fields. We chart topologically equivalent paraxial skyrmion fields which present distinct polarization distributions yet share the same skyrmion number. Doing so not only evidences how standard skyrmion

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structures are connected but, more importantly, allows us to discuss how this polymorphism could be exploited to build novel communication channels taking advantage of the topological nature of optical skyrmions. We generalize our earlier treatment in terms of multiple skyrmion vector potentials [16], to observe degeneracies in constellations of polarization singularity patterns associated with topologically equivalent field configurations.

2. Foundations: a skyrmion model for paraxial light

2.1. Topological considerations and practical realization

Optical paraxial skyrmions are full Poincaré beams [17] characterized by a spatially varying polarization distribution that adheres to strict topological constraints [15]. These are set by the baby Skyrme model [18] from which paraxial skyrmions emerge by mapping the Poincaré sphere onto a S^2 sphere representing a compactified $\mathbb{R}^2$ space. Paraxial skyrmions are therefore constructed according to a two-step process: the set of unitary Stokes vectors $\mathbf{S} = (S_1, S_2, S_3)$ uniquely identifying fully polarized states of light is first anchored into a three-dimensional physical space [15] (step 1 in figure 1). This process associates one polarization state with each point $\mathbf{r} = (x, y, z)$ on S^2 . It is possible, however not essential, to picture these polarization state as unitary vectors $\mathbf{S}'(\mathbf{r}) = (S_x(\mathbf{r}), S_y(\mathbf{r}), S_z(\mathbf{r}))$ in $\mathbb{R}^3$ to enable comparisons with condensed matter skyrmions. Each $\mathbf{r}$, along with its polarization state, are then mapped into the plane transverse to the propagation direction of light, $\mathbb{R}^2$, using a stereographic projection (step 2 in figure 1). This produces a spatially varying polarization distribution, which can be visualized either as a distribution of $\mathbf{S}'(\mathbf{r})$ or as a distribution of polarization ellipses. It is common practice to set $S_x(\mathbf{r}) = S_1(\mathbf{r}); S_y(\mathbf{r}) = S_2(\mathbf{r}); S_z(\mathbf{r}) = S_3(\mathbf{r})$, as for a stereographic projection using the north pole of S^2 as the projection point, the $\mathbf{S}'(\mathbf{r})$ distribution corresponds to a Néel-type (hedgehog) skyrmion. Indeed, using this convention, $\mathbf{S}'$ vectors pointing down at the centre of the transverse plane correspond to left-handed circularly polarized light, $|L\rangle$, since $\mathbf{S}' = \mathbf{S} = (0, 0, -1)$, whereas $\mathbf{S}'$ vectors pointing up at spatial infinity correspond to right-handed circularly polarized light, $|R\rangle$, since $\mathbf{S}' = \mathbf{S} = (0, 0, +1)$. The polarization distribution of a Néel-type skyrmion is illustrated in figure 2(a).

Paraxial skyrmion beams $|\psi(\mathbf{r})\rangle$ are routinely generated by superposing two polarized Laguerre–Gaussian (LG) modes [4]:

$$|\psi(\mathbf{r})\rangle = \frac{1}{\sqrt{2}} \left(\text{LG}_0^{\ell_0} |0\rangle + e^{i\Theta} \text{LG}_0^{\ell_1} |1\rangle \right), \quad (1)$$

where $\ell_{0,1}$ are modal azimuthal indices, $|0\rangle$ and $|1\rangle$ label arbitrary orthogonal polarized states and $e^{i\Theta}$ is an arbitrary phase factor. Bessel modes have recently been proposed as alternatives to LG modes with the aim of generating non-diffracting optical skyrmions beams [19]. The skyrmion number $n = (\ell_1 - \ell_0)$ is an integer counting how many times all polarization states represented on the Poincaré sphere appear in $\mathbb{R}^2$. Equation (1) may be re-written as:

$$|\psi(\mathbf{r})\rangle = \frac{|0\rangle + \mu(\mathbf{r}) |1\rangle}{\sqrt{1 + |\mu(\mathbf{r})|^2}}, \quad (2)$$

where $\mu(\mathbf{r})$ is the ratio between two orthonormal mode functions including $e^{i\Theta}$. Equation (2) shows that the polarization distributions in $\mathbb{R}^2$, hence the polarization distribution on S^2 , is determined solely by $\mu(\mathbf{r})$. Its role is to enforce the topological constraints of the baby Skyrme model as discussed in [15]. $|0\rangle$ and

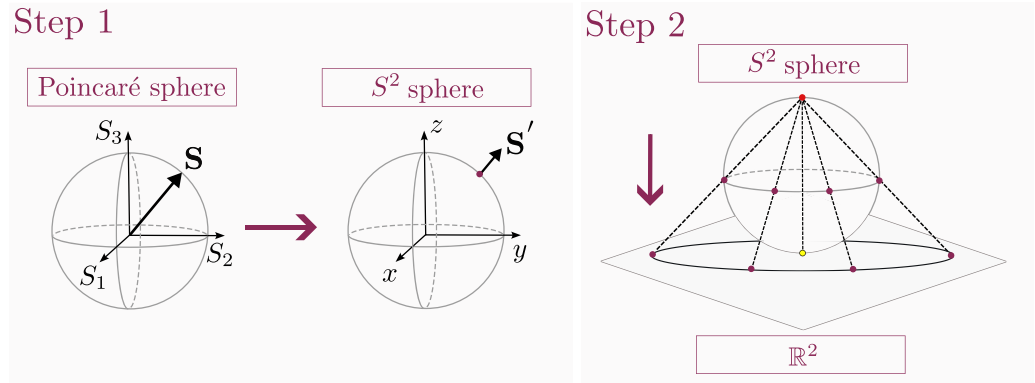


Figure 1. Baby skyrmion model applied to optical paraxial skyrmions. In step 1, unitary stokes vectors $\mathbf{S} = (S_1, S_2, S_3)$ embedded in the abstract space of the Poincaré sphere, are anchored into $\mathbb{R}^3$ yielding $\mathbf{S}'(\mathbf{r}) = (S_x(\mathbf{r}), S_y(\mathbf{r}), S_z(\mathbf{r}))$ according to topological considerations outlined in [15]. In step 2, S^2 is mapped into $\mathbb{R}^2$ using a stereographic projection. Here, the north pole is the projection point by which points $\mathbf{r}$ on S^2 are imaged onto $\mathbb{R}^2$. The imaging process is illustrated for a selection of points on the equator. The south pole is imaged as the origin of $\mathbb{R}^2$ whereas the north pole is imaged at spatial infinity.

$|1\rangle$ determine the polarization states at the poles of S^2 for the stereographic projection shown in figure 1. Setting $\ell_0 = 0$, $\ell_1 = +1$, $|0\rangle = |L\rangle$, $|1\rangle = |R\rangle$, $\Theta = 0$ produces a Néel-type skyrmion of skyrmion number $n = 1$.

2.2. Identifying paraxial skyrmion fields

Skyrmionic beams are readily identified through skyrmion number measurements, where the prevailing method involves measuring the surface coverage of the Poincaré sphere, as described in [2, 4]:

$$n = \frac{1}{4\pi} \int \Sigma_z d\mathbf{A}, \quad \text{with} \quad \Sigma_i = \frac{1}{2} \epsilon_{ijk} \mathbf{S} \cdot \left(\frac{\partial \mathbf{S}}{\partial x_j} \times \frac{\partial \mathbf{S}}{\partial x_k} \right), \quad (3)$$

for a z -propagating beam of light $d\mathbf{A} = dx dy$ is the surface of integration in $\mathbb{R}^2$, Σ is the skyrmion field. A shortfall of this approach is its sensitivity to noise in regions of low intensity, causing n to be depreciated [16]. Stokes' theorem can be invoked to circumvent this difficulty, in which case n becomes [16]:

$$n = \frac{1}{4\pi} \left(\oint_{\mathcal{L}_\infty} \mathbf{V} \cdot d\mathbf{l} - \sum_j \oint_{\mathcal{L}_j} \mathbf{V} \cdot d\mathbf{l} \right) \quad (4)$$

where $\mathbf{V}$ is the skyrmion vector potential constructed as $\Sigma = \nabla \times \mathbf{V}$. $\mathcal{L}_\infty, \mathcal{L}_j$ are circular contours around singular points in $\mathbf{V}$ located at spatial infinity and at j -locations within the beam, respectively. As with the magnetic vector potential in electromagnetism, $\mathbf{V}$ is not uniquely defined and can be transformed using gauge-like transformations [2], this implies that one may examine a Skyrmion field from different perspectives. We can define the vector potential as [2]:

$$\mathbf{V} = -S_z \nabla \Phi \quad \text{where} \quad \Phi = \arctan(S_y/S_x), \quad (5)$$

in which case equation (4) takes yet a simpler form:

$$n = \frac{1}{2} \left(\sum_j S_z^{(j)} N_j - \bar{S}_z^{(\infty)} N_\infty \right) \quad \text{where} \quad N = \frac{1}{2\pi} \oint \nabla \Phi \cdot d\mathbf{l}. \quad (6)$$

Here, $\bar{S}_z^{(\infty)}$ denotes the average value of S_z at spatial infinity. N is an integer counting the number of 0 to 2π phase variations around singular points in the Φ distribution. The skyrmion number can therefore be inferred directly from Φ and, as we shall now see, several phase distributions can be employed to examine a single skyrmion field by changing the correspondence between $\mathbf{S}(\mathbf{r})$ and $\mathbf{S}'(\mathbf{r})$. To that end, we define arbitrary orthogonal polarization states $|\varphi_1\rangle, |\varphi_2\rangle$ in terms of spherical polar coordinates (γ, θ) angles on the Poincaré sphere:

$$|\varphi_1\rangle = \cos\left(\frac{\gamma}{2}\right)|R\rangle + e^{-i\theta}\sin\left(\frac{\gamma}{2}\right)|L\rangle, \quad (7)$$

$$|\varphi_2\rangle = -\sin\left(\frac{\gamma}{2}\right)|R\rangle + e^{-i\theta}\cos\left(\frac{\gamma}{2}\right)|L\rangle. \quad (8)$$

One may then write a general expression for the Pauli matrices:

$$\begin{aligned} \sigma_x &= |\varphi_1\rangle\langle\varphi_2| + |\varphi_2\rangle\langle\varphi_1| \\ &= \begin{pmatrix} 0 & -i\sin\gamma \\ i\sin\gamma & 0 \end{pmatrix} + \cos\gamma \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}, \end{aligned} \quad (9)$$

$$\begin{aligned} \sigma_y &= i(|\varphi_2\rangle\langle\varphi_1| - |\varphi_1\rangle\langle\varphi_2|) \\ &= \begin{pmatrix} \sin\theta & -\cos\theta \\ -\cos\theta & -\sin\theta \end{pmatrix}, \end{aligned} \quad (10)$$

$$\begin{aligned} \sigma_z &= |\varphi_1\rangle\langle\varphi_1| - |\varphi_2\rangle\langle\varphi_2| \\ &= \begin{pmatrix} 0 & i\cos\gamma \\ -i\cos\gamma & 0 \end{pmatrix} + \sin\gamma \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}. \end{aligned} \quad (11)$$

Then, using $\mathbf{S} = \langle\psi(\mathbf{r})|\boldsymbol{\sigma}|\psi(\mathbf{r})\rangle$ we obtain:

$$S_x = \frac{1}{1+|\mu|^2} [2\Re(\mu)\cos\gamma\cos\theta - (1-|\mu|^2)\sin\gamma - 2\Im(\mu)\sin\theta\cos\gamma] \quad (12)$$

$$S_y = \frac{1}{1+|\mu|^2} [2\Re(\mu)\sin\theta + 2\Im(\mu)\cos\theta] \quad (13)$$

$$S_z = \frac{1}{1+|\mu|^2} [2\Re(\mu)\sin\gamma\cos\theta + (1-|\mu|^2)\cos\gamma - 2\Im(\mu)\sin\gamma\sin\theta] \quad (14)$$

where one can verify that $|\mathbf{S}| = 1$. Equations (12)–(14) generalize the expression presented in [2]:

$$S_x = \frac{2\Re(\mu)}{1+|\mu|^2}, \quad S_y = \frac{2\Im(\mu)}{1+|\mu|^2}, \quad S_z = \frac{1-|\mu|^2}{1+|\mu|^2}, \quad (15)$$

which in our general framework correspond to a choice of $\gamma = 0$ and $\theta = 0$. In [2], it is assumed that $S_x(\mathbf{r}) = S_1(\mathbf{r})$, $S_y(\mathbf{r}) = S_2(\mathbf{r})$, $S_z(\mathbf{r}) = S_3(\mathbf{r})$ in equation (15). Using this convention, we can provide a general expression relating $\mathbf{S}(\mathbf{r})$ and $\mathbf{S}'(\mathbf{r})$:

$$S_x = \cos\gamma\cos\theta S_1 - \sin\gamma S_3 - \sin\theta\cos\gamma S_2, \quad (16)$$

$$S_y = \sin\theta S_1 + \cos\theta S_2, \quad (17)$$

$$S_z = \sin\gamma\cos\theta S_1 + \cos\gamma S_3 - \sin\gamma\sin\theta S_2. \quad (18)$$

This enables us to produce equivalent skyrmion potentials for a single skyrmion field simply by choosing different bases $|\varphi_1\rangle, |\varphi_2\rangle$ in σ_i . Figure 2 shows three different Φ distributions corresponding to a given polarization distribution,

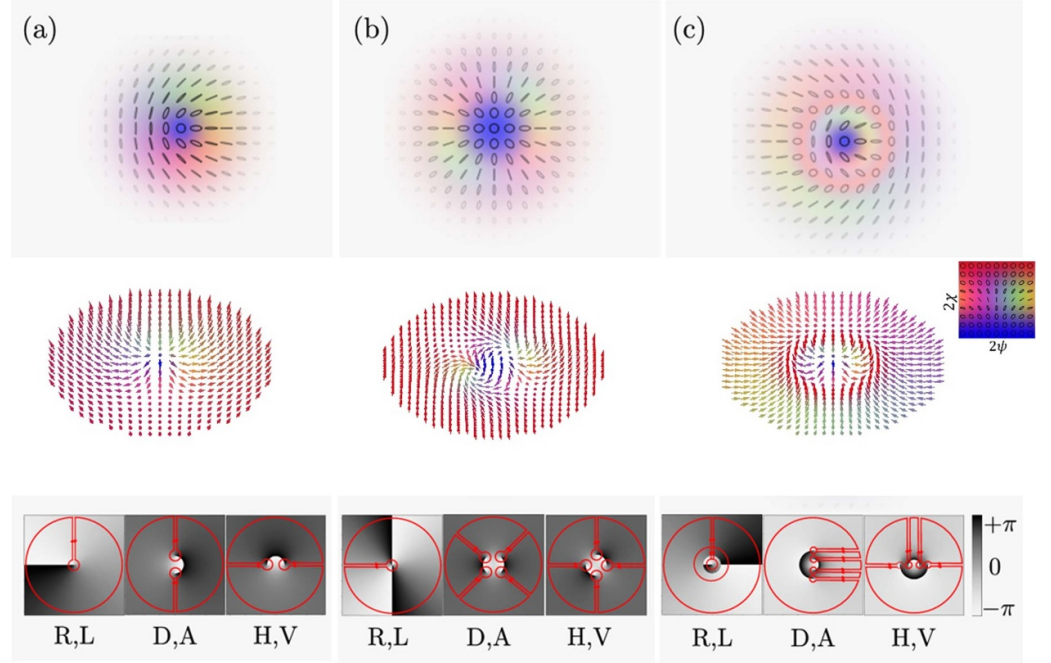


Figure 2. Intensity-modulated polarization distributions for (a) an $n = 1$ and (b) an $n = 2$ Néel-type skyrmion and (c) a general Poincaré beam ($n = 0$). The colour scheme is presented with orientation ψ and ellipticity χ . Three equivalent Φ plots are provided for each polarization distribution, targeting right and left handed circularly polarized light (R), (L), diagonal and antidiagonal linearly polarized light (D), (A) and horizontally and vertically polarized light (H), (V) as phase singularities. Integration contours for equation (4) are indicated in red.

specifically that of a Néel-type skyrmion of skyrmion number $n = 1$ (a), $n = 2$ (b) and of a general Poincaré beam $n = 0$ (c).

The phase $\Phi = \arctan(S_y/S_x)$ is singular when S_y and S_x are both 0. If $(\gamma, \theta) = (0, 0)$, $S_x = S_1$, $S_y = S_2$, $S_z = S_3$ and Φ is singular when $S_2 = S_1 = 0$ which corresponds to C -points, that is, right-handed and left-handed circularly polarized light (R, L plots in figure 2). Similarly, if $(\gamma, \theta) = (-\pi/2, \pi/2)$, we have $S_x = S_3$, $S_y = S_1$, $S_z = S_2$, the phase singularities are diagonal and antidiagonal linearly polarized light (D, A plots in figure 2) whereas choosing $(\gamma, \theta) = (\pi/2, 0)$ gives $S_x = -S_3$, $S_y = S_2$, $S_z = S_1$ corresponding to horizontal and vertical linear polarization states (H, V plots in figure 2).

The choice of triad $\mathbf{S}(\mathbf{r})$ for $\mathbf{S}'(\mathbf{r})$ therefore singles out a pair of orthogonal polarization states to act as phase singularities for Φ . These singularities will occupy different positions within the beam profile but cannot be eliminated in analogy to Dirac strings in the theory of magnetic monopoles [20]. Targeting singularities located in regions of low noise is a strategic choice advantageous for experimental measurements of the skyrmion number, as demonstrated in [16].

3. Generalized skyrmions

3.1. One coverage, countless textures

While Néel-type skyrmions can be regarded as the most fundamental optical skyrmionic structures, one can build an entire atlas of optical skyrmions, encompassing bimerons and Bloch-type skyrmions, by making use of rotations [13, 15]. These do not affect the coverage of the Poincaré sphere hence the skyrmion number, but change the polarization distribution of the beam. Indeed, countless skyrmion structures, all having the same skyrmion number, can be obtained by rotating either the local polarization vectors $\mathbf{S}(\mathbf{r})$ or by rotating

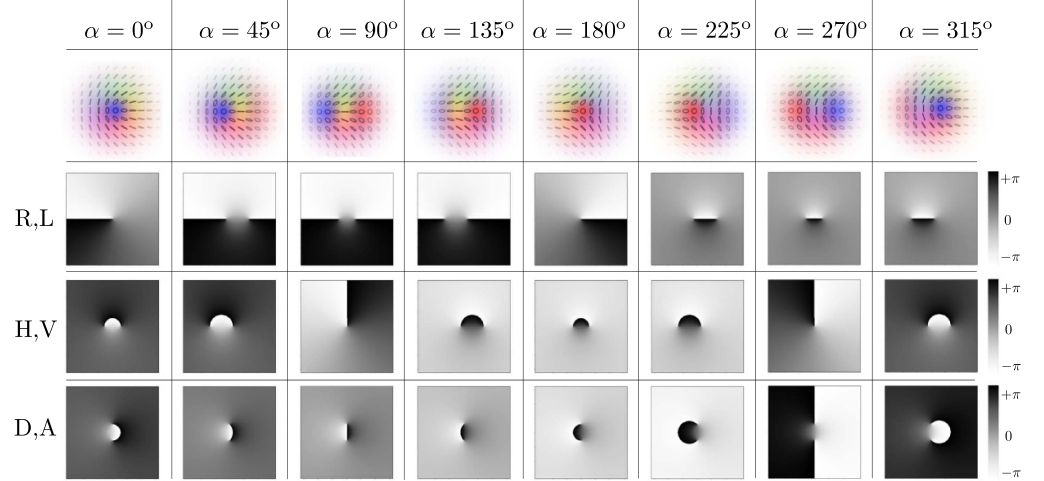


Figure 3. Intensity modulated polarization patterns and phase distributions targeting right and left handed circularly polarized light (R), (L), diagonal and anti-diagonal linearly polarized light (D), (A) and horizontally and vertically polarized light (H), (V) as phase singularities for rotation of $\mathbf{S}$ by an angle α about the S_2 axis, starting from an $n = 1$ Néel-type skyrmion when $\alpha = 0$. All skyrmion beams have a skyrmion number $n = 1$.

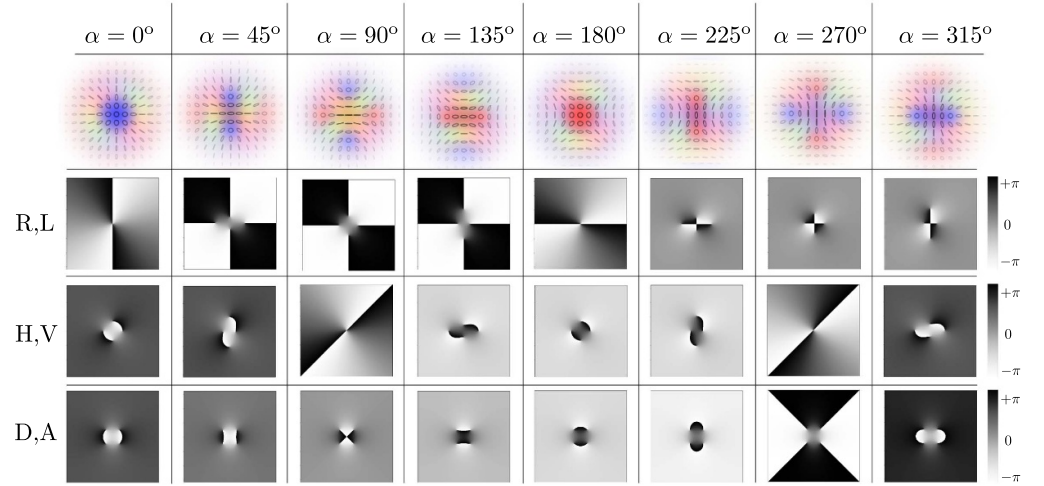


Figure 4. Intensity modulated polarization patterns and phase distributions targeting right and left handed circularly polarized light (R), (L), diagonal and anti-diagonal linearly polarized light (D), (A) and horizontally and vertically polarized light (H), (V) as phase singularities for rotation of $\mathbf{S}$ by an angle α about the S_2 axis, starting from an $n = 2$ Néel-type skyrmion when $\alpha = 0$. All skyrmion beams have a skyrmion number $n = 2$.

S^2 [15]. In equation (1), this freedom is reflected in the choice of polarization states for the orthonormal basis and in the choice of Θ . We show in figures 3 and 4 examples of topologically equivalent optical skyrmions, of skyrmion number $n = 1$ and $n = 2$, respectively, obtained by rotating $\mathbf{S}(\mathbf{r})$ by an angle α about the S_2 axis of the Poincaré sphere. Rotating $\mathbf{S}(\mathbf{r})$ produces a smooth transformation from a Néel-type skyrmion ($\alpha = 0^\circ$ in figure 3) into a bimeron ($\alpha = 90^\circ$ in figure 3). Rotation of S^2 are of different nature, and will transform Néel-type skyrmions into Bloch-type skyrmions while preserving the skyrmion number. Note that these two types of rotations can equally be applied to antiskyrmions, in which case one could effectively create ‘anti-bimerons’ structures. Naturally, figures 3 and 4 only provide eight topologically equivalent textures, but many more can be obtained by considering arbitrary rotations of $\mathbf{S}(\mathbf{r})$, and, for each of these, additional ones can be created by rotating S^2 . This

menagerie of paraxial skyrmions is manifestly vast, and, it would seem that these patterns are completely interchangeable. Based on these considerations, one can produce a set of topologically equivalent skyrmionic structures experimentally by either using $SU(2)$ polarization elements including quarter waveplates and half-waveplates [21] or by rotating the entire polarization profile [4].

3.2. Constellations of polarization singularities

The dynamic evolution of the constellations of polarization singularities observed in the Stokes phase distribution Φ can be studied in figures 3 and 4 as α is varied to create topologically equivalent skyrmion textures. As expected, the choice of basis for Φ affects the trajectory of the individual polarization singularities, with static polarization singularities (here antidiagonal and diagonal linearly polarized states) being obtained when $\mathbf{S}(\mathbf{r})$ is aligned with the rotation axis (here S_2) used to transform the polarization profile. This has implications for the calculation of the skyrmion number using equation (6) as the polarization singularity can migrate to regions of low intensity and deteriorate the measured skyrmion number depending on the choice of the vector potential. Indeed, in our example, it would appear that when $\alpha = 0^\circ$, relying on an H, V basis or a D, A basis to measure the skyrmion number is equally advantageous compared to a measurement using the R, L basis where the singularity of left-handed circularly polarized light is located at the beam periphery. However, the two bases no longer provide the same advantage when the skyrmion beam propagates in a medium that realizes the $SU(2)$ transformation corresponding to a rotation about the S_2 axis as the constellation of H and V singularities move to regions of lower intensity whereas D and A singularities remain in regions of low noise and high-intensity. This is therefore a new aspect to consider in the experimental characterization of skyrmion beams. We also note degeneracies in the Φ patterns across different basis (see for example $R, L, \alpha = 0^\circ$ and $H, V, \alpha = 270^\circ$ in figures 3 and 4). This is expected as the same type of rotation is used to generate different vector potentials and to induce local rotations of the $\mathbf{S}$ vector.

4. Conclusion and outlook

An entire library of topologically equivalent skyrmion fields can be obtained by making use of rotations. We showed using simple examples that, in optics, this resource can readily be accessed experimentally by using simple optical elements. We discussed characterization techniques to identify optical skyrmion beams experimentally, expanding on our previous findings and bringing to light new considerations for best practices. Based on our findings, we conclude that a skyrmion beam propagating in an optically active medium should experience changes in its polarization profile while having its skyrmion number preserved, relevant topics are also discussed in [22]. The skyrmion number should also be conserved when a beam follows a non-planar trajectory [23], making the beam less sensitive to alignment as well as twisting and bending when guided through an optical fibre. Perhaps one of the most promising applications for this rotational invariance lies in the conception of novel communication channels insensitive to misalignment between local reference frames of transmitting and receiving units as discussed in [24] in the context of general rotation-invariant photonic states. Put in perspective with the recent exploration of optical skyrmions in the quantum regime [25], topologically equivalent skyrmion beams could be useful for general bipartite free-space quantum key distribution where information is encoded in the polarization degree of freedom as they can effectively remove the need to establish a shared reference frame [26].

Data availability statement

No new data were created or analysed in this study.

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Conflict of interest

The authors declare no competing financial interests.

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